Master Architectural Blueprint: Enterprise ERP & CRM

Dual-Division Platform Specification: Software Engineering & CCTV Field Operations

This technical architecture blueprint outlines the complete structural implementation for unifying software services and CCTV hardware installations under a single, multi-tenant database. It defines data flow, user roles, mobile field validation, automated billing pipelines, and strict financial controls across all departments.

1. RELATIONAL DATABASE ARCHITECTURE (UNIVERSAL SCHEMA)

Entity Table	Primary & Foreign Keys	Core Attributes & Business Rules
divisions	`id (PK)`	`name ('SOFTWARE', 'CCTV')`. Partitions corporate assets, staff records, and revenue streams.
users	`id (PK), division_id (FK)`	`name, email, role, commission_rate, status`. Enforces RBAC permissions across both companies.
clients & leads	`id (PK), division_id (FK), user_id (FK)`	`company_name, phone, gstin, stage, estimated_value`. Tracks deal lifecycles from intake to closure.
lead_interactions	`id (PK), lead_id (FK), user_id (FK)`	`interaction_type, remarks, follow_up_date`. Chronological ledger of every call, meeting, and remark.
invoices & items	`id (PK), client_id (FK), user_id (FK)`	`subtotal, tax, grand_total, status, serial_numbers`. Captures sales data, deducting stock on issue.
amc_warranties	`id (PK), invoice_id (FK), client_id (FK)`	`service_type, start_date, expiry_date, status`. Drives automated 30/15/7-day renewal workflows.
technician_wallets	`id (PK), technician_id (FK)`	`cash_in_hand, blocked_status`. Live ledger capped at ₹20,000 threshold to prevent unrecorded cash hold.
treasury_ledger	`id (PK), account_id (FK), ref_id (FK)`	`entry_type (DR/CR), amount, running_balance`. Real-time bank and cash ledger updated on all bills/purchases.

2. HIGH-LEVEL ARCHITECTURAL FLOW & PARTITIONING

The system utilizes a centralized backend gateway serving two operational divisions. All incoming REST/WebSocket requests traverse authentication, division verification, and an immutable audit-logging middleware before reaching domain controllers.

Multi-Tenant Division Boundary Rules

1. **Data Isolation:** Client accounts and leads are strictly bound to their respective `division_id`, preventing cross-division access by sales personnel while giving Super Admins unified oversight.
2. **Inventory Duality:** CCTV stock items require physical serial tracking and hardware warranty terms, whereas Software division items utilize recurring subscription licenses or service-milestone deliverables.

3. ROLE-BASED ACCESS CONTROL (RBAC) PERMISSION MATRIX

User visibility and endpoint permissions are enforced via strict JSON Web Tokens (JWT) mapped to granular role definitions.

System Role	Module Access Level	Execution Capabilities & Restrictions
Super Admin	Global (Full Read/Write)	Unrestricted visibility over bank ledgers, audit footprints, staff commissions, and executive live maps.
Accountant	Finance, Billing, Inventory	Generates GST invoices, verifies purchases, settles technician cash wallets, and authorizes expenses.
Sales Representative	CRM, Leads, Catalog	Creates leads, schedules follow-ups, quotes products, and tracks personal commission payouts.
Field Technician	Mobile Tickets, Cash Entry	Closes maintenance orders, logs cash collections, uploads expense proofs; subject to wallet limits.
Field Executive	Mobile CRM, Demo Logging	Restricted mobile view for GPS Check-in/Check-out, lead qualification, and onsite client remark entry.
Software Developer	Sprint Board, Service Tickets	Updates software task states, logs bug resolutions, and manages client onboarding milestones.

4. GLOBAL APPLICATION SHELL ARCHITECTURE

Top Navigation Header

- **Global Entity Search:** Fast indexing for Client Mobile, Invoice ID, or CCTV Serial Number.
- **Notification Tray:** Real-time alerts for expiring AMCs and technicians nearing wallet limits.
- **Division Switcher:** Allows administrators to filter views between Software, CCTV, or Consolidated.

Persistent Sidebar Shell

- **Role-Filtered Navigation:** Dynamically renders route links based on JWT role claims.
- **Live Status Indicators:** Displays active technician alerts and pending approval badges.
- **User Session Profile:** Shows staff name, assigned division, and active token expiration state.

5. THE WORKING FOOTPRINT: TAMPER-PROOF AUDIT MIDDLEWARE

Automated Activity Logging Specification

To ensure total transparency, developers must bind an interceptor middleware to all state-mutating HTTP methods (`POST`, `PUT`, `PATCH`, `DELETE`). Every database modification creates an immutable record inside `activity_logs` storing: `user_id`, `timestamp`, `endpoint`, `ip_address`, and a JSON diff of modified attributes. Deletion of these records is blocked at the database engine level.

6. CRM PIPELINE & FIELD EXECUTIVE GEO-TRACKING

Synchronized lead management connecting central office sales reps with remote field demo staff.

Component Architecture: CRM Lead Engine

A. Frontend UI Layer (React / Next.js)

Interactive Kanban board with stages: *New Lead* → *Contacted* → *Site Demo* → *Quoted* → *Closed Won / Lost*. Drag-and-drop triggers status update APIs. Clicking any card opens a slide-out drawer rendering full chronological call logs, customer profiles, and a mandatory "Next Follow-up" date picker.

B. Backend API Layer & Controllers

`POST /api/leads/interaction` updates stage and sets follow-up reminders. When marked *Closed Won*, the backend returns an action hook prompting the Accountant to issue an invoice with pre-populated client details.

C. Database Transactions

Mutates `leads.stage`, records chronological entry in `lead_interactions`, and writes to `activity_logs`.

Component Architecture: Field Executive Geo-Verification

A. Mobile Web Interface (Geolocation API)

Mobile-optimized dashboard displaying daily assigned visit routes. Features prominent **Start Visit (GPS Check-In)** and **End Visit (GPS Check-Out)** buttons, paired with client visit proof photo upload.

B. Anti-Fraud Backend Logic

The backend captures device coordinates at start and finish, running a Haversine distance algorithm against the client's registered address. If the radius exceeds 150 meters, the visit is flagged for managerial review. Check-Out requires entering mandatory demo notes.

C. Database Records

Stores check-in/out timestamps, GPS coordinates, meeting duration, and photo URLs in `field_visits`.



7. BILLING, INVENTORY & AUTOMATED SALES COMMISSIONS

Integrated invoicing engine that dynamically decrements warehouse stock and calculates employee sales commissions.

Invoicing & Commission Pipeline Architecture

A. Frontend UI Layer

Dual billing interface allowing selection of software service packages or CCTV physical hardware. Hardware lines require scanning or typing exact serial numbers. Includes one-click PDF invoice rendering and print formatting.

B. Backend Business Logic & Event Handlers

1. **Stock Deduction:** On invoice submission, CCTV item inventory counts decrease by invoiced quantities.
2. **Commission Trigger:** When an invoice transitions to `PAID`, a background worker identifies the assigned sales representative, applies their commission percentage, and records an approved payout in `commissions`.

C. Database Entities Affected

`invoices`, `invoice_items`, `products_inventory (stock_qty)`, and `commissions`.

8. AMC & WARRANTY AUTOMATION ENGINE

Preventive Expiry & Renewal Alert Architecture

A. Frontend Dashboard Matrix

Color-coded triage screen showing active warranty contracts. Features filters for expiring assets, direct customer contact triggers, and a one-click button to generate an AMC renewal quote pre-filled with client history.

B. Midnight Background CRON Engine

A scheduled script executes every night at 00:00 UTC querying `amc_warranties` for records where `expiry_date - CURRENT_DATE` equals 30, 15, or 7 days. The engine dispatches automated email/SMS reminders to clients while alerting the assigned account manager.

Contract Status	Trigger Condition	Automated System Action	Target Recipient
ACTIVE	Expiry > 30 Days	Normal status; service tickets enabled.	Dashboard View
WARNING 30D	Expiry = 30 Days	Generates advance renewal notification.	Client & Sales Rep
URGENT 15D	Expiry = 15 Days	Elevates priority; flags row in yellow on UI.	Accountant & Sales
CRITICAL 7D	Expiry ≤ 7 Days	Dispatches final alert and quote link.	Client & Super Admin
EXPIRED	Expiry < 0 Days	Locks free service calls; flags for paid AMC.	Technician Portal

9. TECHNICIAN CASH WALLET & TREASURY FLOW CONTROLS

Operational financial safety controls to prevent unrecorded cash retention and track company cash flow.

Technician Cash Control & The ₹20,000 Hard Limit

A. Mobile Cash Receipting & Real-Time Customer Verification

When a technician completes an installation and receives cash, they enter the exact amount into the mobile app to close the ticket. The system sends an automated SMS/WhatsApp receipt to the client, preventing under-reporting.

B. The ₹20k Ticket Block Engine

Logged collections immediately increment `technician_wallets.cash_in_hand`. If this balance reaches or exceeds ₹20,000, the system automatically revokes the technician's ability to accept or view new work tickets until the Accountant formally logs cash settlement.

C. Daily Expense Claims (Petrol, Hardware, Consumables)

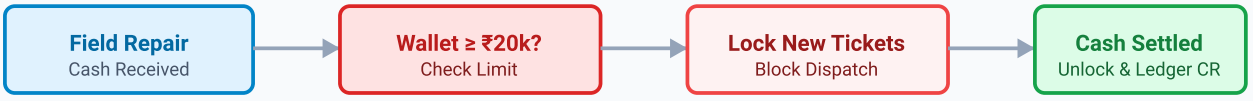
Technicians upload same-day expense requests with photos of receipts. When approved by the Accountant, the amount is deducted from their cash-in-hand wallet balance, lowering their liability.

10. REAL-TIME TREASURY & DOUBLE-ENTRY CASH FLOW

Live Bank & Petty Cash Ledger Integration

Automated Inbound & Outbound Balancing

Every business action mutates the `treasury_ledger`: customer invoice payments credit company bank accounts; hardware stock purchases debit accounts; technician wallet handovers credit the petty cash ledger. This provides Super Admins with real-time liquidity reporting.



11. IMPLEMENTATION ROADMAP

- **Phase 1: Foundation (Weeks 1–2):** Set up PostgreSQL schema, JWT authentication, and the immutable audit middleware.
- **Phase 2: CRM & Field Geo-Tracking (Weeks 3–5):** Build the Kanban pipeline and mobile GPS Check-in/Check-out system.
- **Phase 3: Billing, AMC & Wallets (Weeks 6–8):** Deploy invoice generation, the midnight AMC CRON, and the ₹20,000 wallet limit block.
- **Phase 4: Treasury & Reporting (Weeks 9–10):** Finalize double-entry bank tracking, staff commission aggregation, and analytics.